AI-Powered Tour Guide Booking System with Voice and Retrieval Augmented Generation

(24-25J-150)

PRAVEAN P.R.I

(IT21191060)

B.Sc. (Hons) Information Technology in Information Systems Engineering

Department of Computer Systems Engineering

Sri Lanka Institute of Information Technology

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Project Proposal Report

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Declaration

I declare that this is our own work, and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

Name	Student ID	Signature
P.R.I. PRAVEAN	IT21191060	

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

.....

Signature of the supervisor:

(Ms. Suranjini Silva) Date: 27/09/2025

.....

Signature of the Co-supervisor:

Date: 27/09/2025

Abstract

Tourism plays a vital role in the Sri Lankan economy, attracting millions of visitors each year who seek its cultural heritage, natural beauty, and diverse experiences. However, one of the major pain points faced by tourists is the difficulty in finding trustworthy, safe, and suitable tour guides. Existing online platforms such as TripAdvisor, Viator, and GetYourGuide provide only basic functionalities like listings, reviews, and ratings. These systems often fall short in three critical areas: verification, personalization, and safety. Tourists continue to face challenges such as unverified guides, misleading reviews, mismatched services, and premature exposure of personal details. This lack of reliability reduces trust in digital booking systems and can compromise the overall tourist experience, as well as the reputation of the destination.

To address these shortcomings, this research proposes the development of an AI-Powered Tour Guide Booking System with Voice and Retrieval-Augmented Generation (RAG). The platform is designed as a secure, intelligent, and user-friendly web application that combines multi-level verification, AI-driven personalization, and privacy-preserving workflows. Unlike traditional booking systems, our solution ensures that guides undergo rigorous verification processes using multiple documents, including government IDs, licenses, police checks, and optional social media validation. This provides tourists with confidence that they are interacting with legitimate, trusted guides.

The system also integrates advanced AI-based matching and recommendation features, powered by RAG, to deliver personalized guide suggestions that go beyond static ratings. By analyzing tourist preferences, cultural contexts, past booking history, and even real-time data such as weather or festivals, the platform ensures a truly tailored experience. In addition, a multilingual AI chat agent, supporting Sinhala, Tamil, and English, enables tourists to make inquiries in their preferred language via voice or text, ensuring accessibility across diverse audiences. The system's scalability allows it to be expanded beyond Sri Lanka, positioning it as a global digital tourism innovation. By combining technological advancements with the core values of safety, trust, and personalization, the solution not only enhances individual travel experiences but also contributes to strengthening Sri Lanka's reputation as a reliable and smart travel destination.

In summary, this project introduces a next-generation tourism platform that bridges the gap between tourists' expectations and the limitations of current systems. By integrating AI, RAG, and privacy-focused workflows into a single solution, the AI-Powered Tour Guide Booking System provides a balanced trust model where safety, personalization, and user satisfaction are at the forefront. This ensures that every journey is not only memorable but also secure, setting a new standard for the future of digital tourism.

List of Abbreviations

LLM	Large Language Model
DB	Database
SQL	Structured Query Language
AI	Artificial Intelligence
UI/UX	User Interface and User Experience
GDPR	General Data Protection Regulation
API	Application Programming Interface
RAG	Retrieval Augmented Generation

Table 1: List of Abbreviation

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1.Introduction

1.1 Research Gap

Tourism is one of the fastest-growing sectors in the global economy and holds particular significance for Sri Lanka, a country renowned for its natural beauty, cultural heritage, and hospitality. Each year, millions of visitors arrive to experience the country's unique blend of beaches, wildlife, history, and traditions [1]. Beyond its cultural and social importance, tourism directly contributes to economic growth, job creation, and foreign exchange earnings. However, despite these benefits, the tourism industry continues to face persistent challenges that impact both travelers and service providers [2]. One of the most critical challenges is ensuring that tourists can find reliable, safe, and verified tour guides who align with their preferences, language needs, and budget.

Tourists often encounter unverified guides, misleading reviews, or incomplete information, which can lead to negative experiences or even safety concerns. Additionally, the absence of strong privacy measures results in premature exposure of personal details, putting tourists at risk of fraud or misuse of information. Another limitation is the lack of intelligent personalization existing systems primarily rely on static ratings and categories rather than adapting to dynamic user preferences, cultural contexts, or real-time conditions such as weather, local events, or emergencies.
The consequences of these shortcomings are far-reaching. For tourists, it reduces trust and creates anxiety about whether their chosen guide will deliver a safe and satisfying experience. For guides, it limits opportunities to showcase their skills and credibility, as existing systems fail to differentiate between verified professionals and unverified individuals. For Sri Lanka's tourism industry, these issues harm reputation, reduce visitor satisfaction, and hinder long-term sustainability [3].

To address these gaps, this project proposes the development of an AI-Powered Tour Guide Booking System with Voice and Retrieval-Augmented Generation (RAG). The system is envisioned as a secure, intelligent, and user-friendly web platform that integrates multi-document verification, AI-driven personalization, and privacy-preserving workflows to deliver a new standard of digital tourism services. Unlike conventional booking systems, this platform categorizes guides as verified or non-verified, ensuring transparency for tourists. Verification processes may include government-issued identification, police checks, professional licenses, and optional social media validation.

Beyond verification, the platform introduces advanced AI matching and recommendation features. With the integration of RAG, tourists can make voice or text-based queries, and the system will respond with context-aware, multilingual, and personalized suggestions. This capability ensures that tourists are paired with guides who best meet their cultural, linguistic, and experiential needs. The solution also incorporates fraud detection mechanisms, emergency response tools, and post-trip feedback systems, thereby addressing both safety and service quality concerns.

Ultimately, the goal of this project is to bridge the gap between technology and trust in tourism. By leveraging modern web technologies, artificial intelligence, and secure workflows, the proposed system not only improves the tourist experience but also empowers guides and enhances Sri Lanka's global image as a safe, innovative, and tourist-friendly destination.

2. Research Problem

The global tourism industry has seen a rapid digital transformation over the past two decades, with platforms such as TripAdvisor, Viator, GetYourGuide, Airbnb and Experiences dominating the market for activity and guide bookings. These systems have undoubtedly enhanced accessibility and convenience for travelers, but they continue to exhibit critical gaps that limit trust, safety, and personalization. Understanding these gaps is essential in defining the need for a more advanced, AI-driven solution tailored to modern tourism challenges.

One of the most significant gaps is the lack of robust verification mechanisms. Current platforms primarily rely on self-reported information, reviews, and ratings to build trust between tourists and guides. While some basic verification (such as email or ID checks) exists, it is often superficial and does not involve multi-document validation or background checks. As a result, tourists face the risk of engaging with unverified or fraudulent guides, leading to safety issues, scams, or poor-quality services. This gap in authentication directly undermines the reliability of existing booking systems [3].

Another notable shortcoming is the personalization of services. Most platforms categorize guides and activities into broad groups and rank them based on static ratings. However, they rarely consider real-time contextual factors such as festivals, weather conditions, or emerging safety advisories. Moreover, they do not dynamically adapt to tourists' cultural backgrounds, language preferences, or historical booking behavior. This leads to mismatched services that fail to deliver meaningful or memorable travel experiences. The lack of intelligent recommendation engines that can understand nuanced preferences highlights a key research gap in personalization.

In many existing platforms, tourists' personal details are shared with guides or third parties before bookings are confirmed. This premature exposure creates risks of data misuse, spam, and even identity fraud. Privacy-preserving workflows that conditionally unlock sensitive details only after a booking is finalized are largely absent in current systems, exposing both tourists and guides to vulnerabilities [4][5].

Furthermore, existing platforms offer little to no built-in fraud detection or emergency support. Tourists are often left on their own if they encounter scams, unsafe situations, or emergencies during trips. While reviews and ratings provide some retrospective accountability, they do not prevent incidents in real time. The absence of integrated fraud detection algorithms, predictive alerts, or location-based emergency features creates a major gap in ensuring traveler safety and trust [1].

Finally, there is limited support for multilingual and culturally adaptive interactions. Most platforms operate predominantly in English or a few major languages, creating accessibility barriers for both tourists and local guides in diverse regions like Sri Lanka. Without multilingual AI-driven assistance, many users are excluded from engaging effectively with digital platforms.

In summary, the existing body of research and practice in digital tourism platforms reveals clear gaps in verification, personalization, privacy, fraud detection, safety, and accessibility. While

current solutions emphasize convenience and scalability, they fall short in fostering trust and security, two critical pillars of tourism. This research project directly addresses these gaps by proposing an AI-powered tour guide booking system with voice and Retrieval-Augmented Generation (RAG) that integrates multi-document verification, personalized AI matching, multilingual support, fraud detection, and privacy-aware workflows. By filling these gaps, the project aims to redefine the standard for secure and intelligent digital tourism.

	Booking & Reviews	Tour Plan Matching	Verified guides
TripAdvisor	✓	✓	✗
GetYourGuide	✓	✗	✗
Viator	✓	✗	✗

Figure 1: Competitive Analysis

3.Objectives

3.1. Main Objective

The primary aim of this project is to develop a secure, intelligent, and user-friendly tour guide booking system that integrates artificial intelligence, multi-document verification, and privacy-preserving mechanisms to address the current limitations in digital tourism platforms. By focusing on trust, personalization, and safety, the project seeks to create a balanced solution that benefits tourists, guides, and the broader tourism industry in Sri Lanka and beyond.

Main Objective of the project is to develop a secure web application for tour guide bookings that incorporates multi-level verification, privacy-protected detail unlocking, and AI-assisted matching and recommendations, thereby improving trust, safety, and efficiency in tourism services.

This objective directly responds to the key issues identified in the research gap, namely the absence of strong verification, intelligent matching, and built-in safety mechanisms in existing booking systems.

3.2 Specific Objectives

The project outlines the following specific objectives that will guide the overall development process:

3.2.1. To implement a multi-document verification workflow that ensures guides are categorized as verified or non-verified based on identity documents, professional licenses, police checks, and optional social media validation. This objective enhances trust and authenticity in the platform.

3.2.2 To develop an AI-powered recommendation and matching engine using Retrieval-Augmented Generation (RAG) to provide personalized guide suggestions based on tourist preferences, cultural contexts, booking history, and real-time data such as weather, festivals, and safety advisories.

These objectives address the dual challenges of security and personalization, which are essential for building a trusted and intelligent booking ecosystem.

3.3 Expected Outcome of Objectives

By achieving the above objectives, the project will deliver a platform that:

3.3.1 Build trust through strong verification and fraud prevention.
3.3.2 Delivers personalized experiences via AI-driven recommendations.
3.3.3 Protect privacy through conditional detail sharing.
3.3.4 Ensures safety with emergency response systems.
3.3.5 Promotes accessibility through multilingual and voice-enabled interactions.

In conclusion, the objectives of this project are carefully designed to bridge the research gaps identified earlier. They provide a clear roadmap for developing an innovative, secure, and intelligent solution that has the potential to redefine digital tourism services, particularly in Sri Lanka's context.

4.Methodology

The methodology for this project is designed to ensure a structured, research-driven, and iterative approach to developing the proposed AI-Powered Tour Guide Booking System with Voice and RAG. It involves three key stages: requirement gathering, past research analysis, and the application of an appropriate development methodology. Together, these steps provide a comprehensive framework for ensuring that the final solution is robust, reliable, and aligned with both technical and tourism industry needs.

4.1 Requirement Gathering

Requirement gathering is the foundation of this project, as it ensures that the proposed system addresses the real needs of tourists, guides, and other stakeholders in the tourism sector. Information will be collected through multiple approaches:

4.1.1 Stakeholder Interviews: Discussions with tourists, tour guides, and tourism board officials to identify pain points such as lack of verification, privacy concerns, or mismatched services.

4.1.2 Competitor Analysis: Evaluation of existing platforms like TripAdvisor, Viator, and GetYourGuide to identify gaps in verification processes, personalization, fraud detection, and emergency support.

4.1.3 Surveys and Questionnaires: Collecting data from potential users to understand preferences, language needs, and trust factors influencing booking decisions.

4.1.4 Regulatory Review: Assessing tourism and regulations (e.g., GDPR, Sri Lankan tourism authority requirements) to ensure compliance.

The outcome of this stage will be a detailed requirements specification that defines the functional and non-functional aspects of the system.

4.2. Past Research Analysis

To build on existing knowledge, the project incorporates insights from academic research and industry case studies related to digital tourism, AI-powered recommendation systems, and cybersecurity in peer-to-peer platforms. Key findings from past research include:

4.2.1 The limitations of review-based trust models, which are often subject to manipulation.

4.2.2 The importance of multi-layer verification to prevent fraud and ensure safety.

4.2.3 The growing role of artificial intelligence in personalization, where RAG and NLP can significantly improve user satisfaction by providing context-aware recommendations.

4.2.4 The lack of integrated emergency and fraud detection systems in current digital tourism solutions. By analyzing this prior work, the project identifies clear gaps and validates the novelty of its proposed solution. This stage ensures that the system is grounded in research while innovating beyond existing limitations.

4.3 Development Methodology

Given the dynamic and user-centric nature of the project, the Agile development methodology will be adopted. Agile emphasizes iterative development, flexibility, and continuous feedback, making it ideal for building and refining a complex AI-driven web platform. The key steps include:

4.3.1 Prototype Development: Creating early prototypes of the user interface, verification module, and AI chat agent for testing with stakeholders.

4.3.2 Incremental Development: Building the system module by module, including verification workflows, booking engine, AI matching, fraud detection, and emergency features.

4.3.3 Testing and Feedback: Conducting usability testing, security audits, and multilingual validation with real users during each sprint.

4.3.4 Deployment and Evaluation: Launching a pilot version for beta testing and refining the platform based on performance metrics such as booking success rate, AI accuracy, and user satisfaction.

This methodology ensures that the project remains user-focused, adaptive to feedback, and aligned with the overall objectives of improving trust, safety, and personalization in tourism services.

4.4 Project Management Methodology

Trello and MS planer will be used for the planning, tracking, and management of the loan prediction platform development. We will be able to organize tasks, assign responsibilities, and monitor progress using Jira throughout the life cycle of the project.

5. Feasibility Study

The feasibility study evaluates whether the proposed AI-Powered Tour Guide Booking System with Voice and RAG can be successfully developed and deployed. This analysis considers technical readiness, scheduling practicality, economic viability, and the availability of tools and knowledge to execute the project effectively.

5.1 Technical Feasibility

The project is technically feasible due to the availability of mature frameworks, AI libraries, and cloud platforms that support scalable web applications. The frontend will be developed using React.js, ensuring interactive and responsive interfaces. The backend will utilize Node.js/Express, which is well-suited for handling API integrations and real-time data flows. For intelligent recommendations, Retrieval-Augmented Generation (RAG) combined with NLP models will be used to match guides to tourists based on preferences and contextual data.

A vector database such as Pinecone, FAISS, or Weaviate will support efficient information retrieval. On the security side, the system will employ OAuth 2.0 authentication, HTTPS protocols, and database encryption to protect sensitive user information. These technologies are widely available, proven, and supported by extensive developer communities, making implementation realistic and reliable

5.2 Schedule Feasibility

The project is designed to be completed within 6–8 months, divided into multiple phases using the Agile methodology.

- In Months 1 to 2: Requirement gathering, stakeholder interviews, competitor analysis, and system design.
- In months 3 to 4: Development of core modules such as guide verification workflows, booking system, and AI-based matching engine.
- In months 5 to 6: Integration of chat agent, fraud detection, multilingual support, and emergency features.
- In month 7 Security testing, usability validation, and pilot deployment.
- In month 8 Evaluation and refinement.

This schedule ensures that all functional and non-functional requirements are addressed while leaving room for iterative improvements based on testing feedback.

5.3 Economic Feasibility

From a financial standpoint, the project is viable with a moderate level of investment. Initial costs will cover platform development, security integrations, AI model training, and API subscriptions for verification services. Estimated costs are as follows:

- Development and Deployment: RS 20,000 to RS 25,000
- Security & Verification APIs: RS 5,000
- Maintenance (annual): RS 5,000

Revenue generation will rely on a commission-based model (10–15% per booking), subscription plans for premium guide profiles, and partnerships with hotels and travel agencies. With steady adoption, the system is expected to achieve break-even within 2–3 years, making it economically sustainable.

6. System Analysis

The Software Solution Approach for the proposed AI-Powered Tour Guide Booking System with Voice and RAG emphasizes a systematic and iterative development process. It ensures that the final system is not only technologically robust but also user-friendly, secure, and aligned with the real-world needs of tourists and guides. The approach is divided into four main stages: requirement gathering, interactive modules and tools development, user experience and interface design, and testing and integration.

6.1. Requirement Gathering

The process begins with comprehensive requirements gathering to ensure that the platform addresses the needs of all stakeholders, including tourists, tour guides, and regulatory authorities. This involves:

6.1.1 Conducting interviews and surveys with tourists to identify expectations for safety, language support, and personalization.

6.1.2 Incremental Development: Building the system module by module, including verification workflows, booking engine, AI matching, fraud detection, and emergency features.

6.1.3 Testing and Feedback: Conducting usability testing, security audits, and multilingual validation with real users during each sprint.

6.1.4 Deployment and Evaluation: Launching a pilot version for beta testing and refining the platform based on performance metrics such as booking success rate, AI accuracy, and user satisfaction.

The outcome of this stage will be a comprehensive Software Requirement Specification (SRS), outlining functional, non-functional, and regulatory requirements.

6.2. Interactive Modules and Tools Development

The system will be built around several core modules, each developed incrementally and integrated into the platform:

6.2.1 Guide Verification Module – Multi-document verification (ID, license, police check, social media linking).

6.2.2 Booking & Matching Engine – AI-powered RAG-based recommendation system for tourist–guide pairing.

6.2.3 Privacy and Safety Module – Conditional data unlocking, emergency alerts, location tracking.

6.2.4 Multilingual AI Chat agent – Voice/text-based interactions in English, Sinhala, and Tamil for inquiries, booking, and safety guidance.

6.2.5 Fraud Detection Engine – Real-time monitoring and flagging of suspicious activity using AI models.

Each module will be developed with modern tools such as React.js for frontend, Node.js/Express for backend, Lang Chain for AI orchestration, and vector databases for efficient retrieval in RAG-powered queries.

6.3 User Experience and Interface Design

Given that the platform is intended for global tourists with diverse backgrounds, UX/UI design is a critical focus area. The interface will be:

6.3.1 Simple and intuitive, enabling tourists to search and book guides with minimal effort.
6.3.2 Privacy-focused, with clear indicators on when personal data is locked or shared.
6.3.3 Multilingual and voice-enabled, making the system accessible to users with different language preferences.
6.3.4 Mobile-responsive, ensuring functionality across devices including smartphones, tablets, and desktops.

Prototyping tools from Figma will be used to design wireframes and mockups, which will undergo usability testing with real users before full implementation.

6.4 Testing and Integration

The final stage emphasizes rigorous validation to ensure system reliability and trustworthiness:

6.4.1 Unit and Integration Testing for each module to confirm functional correctness.
6.4.2 Security Testing includes penetration testing and data encryption checks to protect sensitive information.
6.4.3 Usability Testing with tourists and guides to refine workflows and improve satisfaction.
6.4.4 System Integration Testing to ensure seamless communication between modules such as booking, verification, and AI Chat agent.
6.4.5 Pilot Deployment in a limited region (e.g., Colombo or Anuradhapura) to evaluate system performance in real-world conditions before scaling nationwide.

By following this structured approach, the software solution ensures a balance of security, personalization, usability, and scalability, addressing the shortcomings of existing tourism platforms and setting a new benchmark in intelligent digital tourism.

7. System Overall Diagram

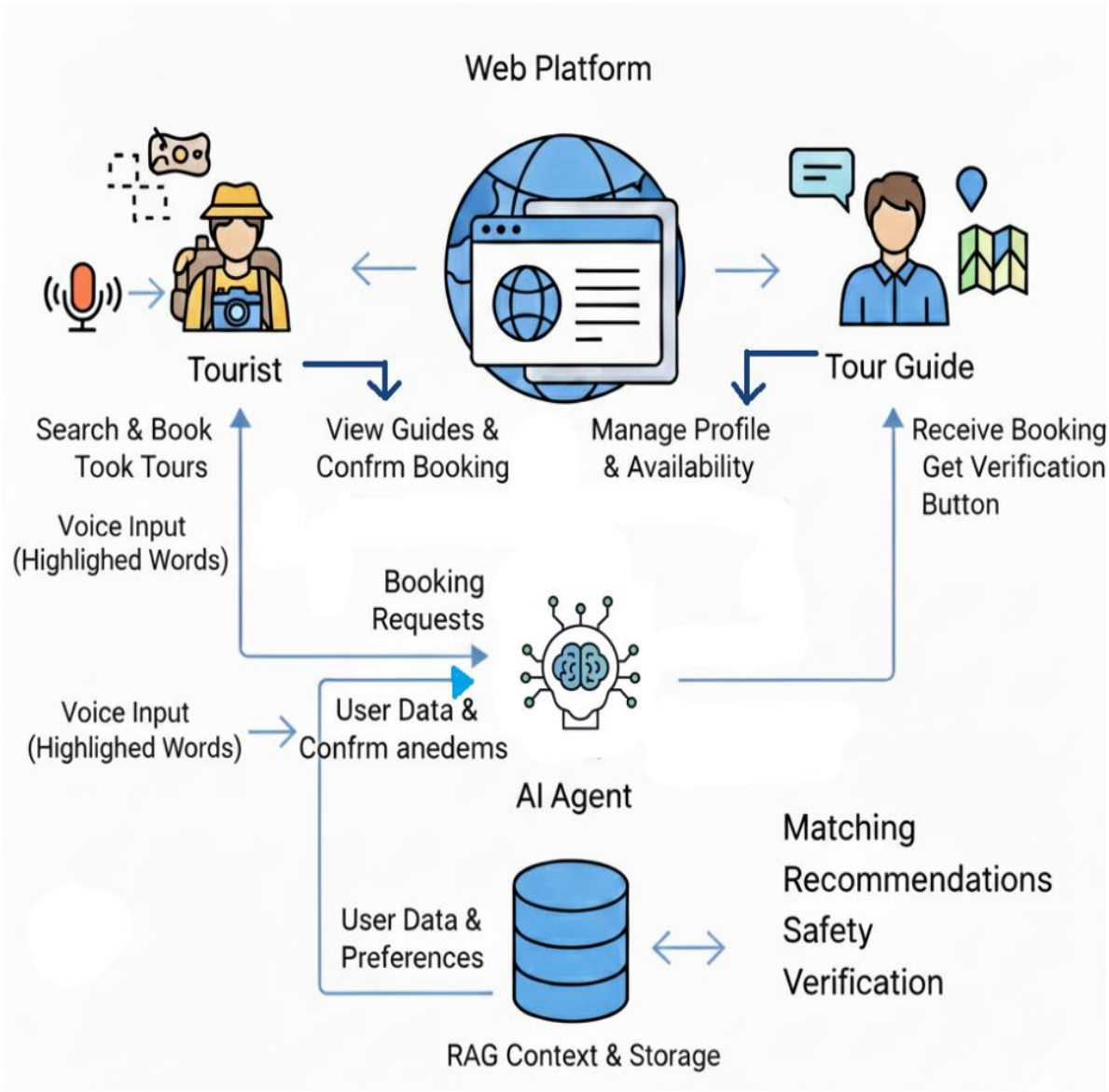

Figure 2: System Diagram

8. Work Breakdown Structure

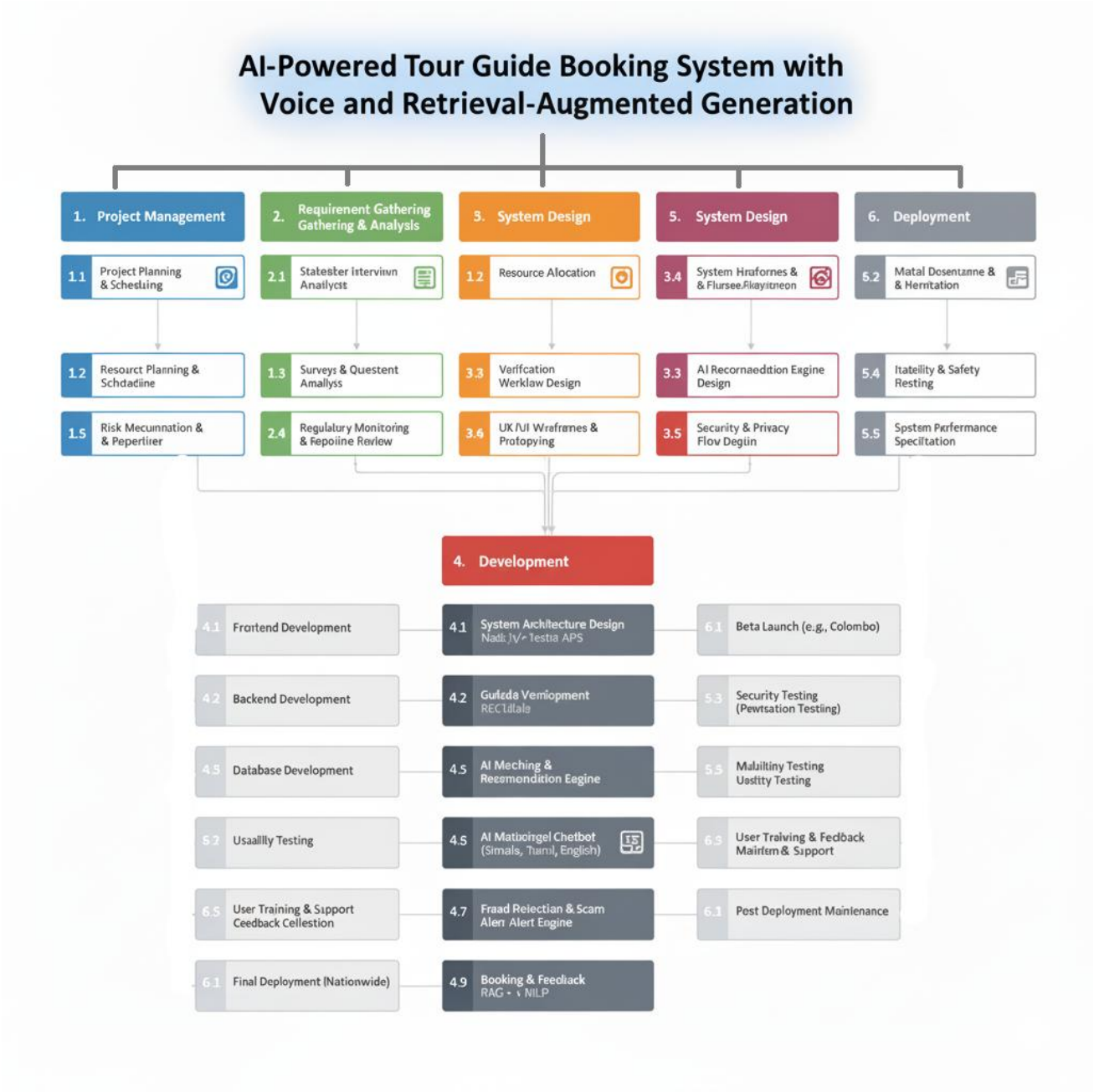

Figure 3: Work Breakdown Structure

9. Tools & Technology

Frontend Development

- React.js – Interactive and responsive user interfaces
- HTML5, CSS3, JavaScript Core web technologies
- Figma for prototyping

Backend Development

- Node.js with Express.js – Server-side logic and RESTful APIs
- PostgreSQL / MySQL – Relational database for booking and user data
- NoSQL (MongoDB) – Flexible storage for unstructured data

AI & Machine Learning

- RAG (Retrieval-Augmented Generation) – Personalized tourist–guide matching and recommendations
- LangChain / LlamaIndex – Orchestration of AI workflows
- OpenAI GPT / Hugging Face Models – NLP for Chat agent, keyword extraction, and multilingual support
- Vector Databases (Pinecone, Weaviate, FAISS) – Efficient retrieval for AI recommendations
- OCR Tools (Tesseract) – Document validation for guide verification

10. Project Requirements

A clear definition of project requirements and scope ensures that the proposed AI-Powered Tour Guide Booking System with Voice and RAG is developed with well-structured goals, realistic boundaries, and measurable deliverables. These requirements are classified into functional and non-functional categories, while the project scope defines what will and will not be included in the solution.

10.1 Functional Requirements

Functional requirements describe the core features and operations the system must perform:

10.1.1 User Registration and Authentication – Tourists and guides must register securely, with guides uploading documents for verification.
10.1.2 Multi-Document Verification – Validation of NIC, passport, licenses, and police check to categorize guides as verified or non-verified.
10.1.3 Tourist–Guide Matching – AI-powered RAG-based engine to recommend guides based on tourist preferences, location, and real-time factors.
10.1.4 Privacy-Aware Booking Flow – Tourists' personal details remain locked until a booking is confirmed.
10.1.5 Multilingual AI Chat agent – Voice and text-enabled assistant supporting Sinhala, Tamil, and English for bookings, recommendations, and inquiries.
10.1.6 Fraud Detection and Scam Alerts – AI-driven monitoring and alerts for suspicious behavior.
10.1.7 Feedback and Rating System – Post-trip reviews to improve trust and future recommendations.

10.2 Non-Functional Requirements

These define the system's performance, usability, and compliance standards:

10.2.1 Security – OAuth 2.0 authentication, and GDPR compliance for data protection.
10.2.2 Scalability – The system must support increasing numbers of tourists and guides without performance degradation.
10.2.3 Performance – Response time for AI queries and bookings should be less than 2 seconds.
10.2.4 Reliability – guaranteed through cloud deployment and redundancy.
10.2.5 Cross-Platform Accessibility – Full responsiveness across mobile, tablet, and desktop devices.
10.2.6 Usability – Interfaces must be intuitive and user-friendly, requiring minimal learning effort.

11. Project Scope

11.1 Project In-Scope

The project scope outlines what will be covered in the system development and what will be excluded to maintain focus.

- Secure web platform with tourist and guide registration.
- Multi-level guide verification and tiered categorization.
- AI-powered matching and recommendation engine using RAG.
- Multilingual Chat agent with text and voice query support.
- Fraud detection, scam alerts, and emergency response features.
- Privacy-aware workflows for conditional unlocking of tourist details.
- Post-trip feedback and rating system for continuous improvement.

11.2 Project Out of Scope

- Airline and hotel booking integrations (to remain focused on guides).
- Offline or cash-based payment handling (initial version will focus on digital transactions).
- Augmented Reality (AR) or Virtual Reality (VR) guided tours (may be considered as future enhancements).
- Insurance and visa services (to be handled by third-party providers).

12. Timeline

The proposed timeline for the project is as follows.

Figure 4: Grantt Chart

13. Commercialization

The commercialization strategy for the AI-Powered Tour Guide Booking System with Voice and RAG is designed to ensure both sustainability and scalability. The platform will initially launch in Sri Lanka, targeting popular tourist destinations such as Colombo, Kandy, Galle, and Anuradhapura. Once validated locally, the system can be expanded regionally and globally, offering a strong competitive edge in the digital tourism market.

Revenue will primarily be generated through three models:

- Commission-Based Revenue:
  A 10% to 15% commission charged on each successful booking between tourists and guides.
- Subscription Plans for Guides:
  Premium accounts offering enhanced visibility, advanced analytics, and featured listings for verified guides.
- Partnerships and Collaborations:
  Strategic partnerships with hotels, airlines, and travel agencies to integrate the system into tourism packages.

Future opportunities include expanding into value-added services such as travel insurance, curated experiences, and AI-based itinerary planning. The system's scalability, strong verification model, and emphasis on privacy and safety position as a trusted global tourism solution, ensuring long-term profitability.

14. Budget

The deliverable of the proposed model is a software-based solution, there are no Hardware components connected to the implementation. The biggest source of the cost will be the personnel costs, hardware and software for the computing power of the machine system. However, there will be some other costs expected that will be given in the table below.

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Type	Cost (LKR)
Personnel Costs	70 000
Model Development and Training	50 000
Integration and Deployment	60 000
Project Management and Administration	20 000
Total Project Budget	200 000

Table 2: Cost Management Plan

15. Summary

Tourism is a vital driver of Sri Lanka's economy, yet tourists often face challenges in finding reliable, safe, and verified tour guides. Existing platforms such as TripAdvisor and Viator offer convenience but lack multi-level verification, robust fraud detection, privacy-aware workflows, and intelligent personalization. These shortcomings reduce trust, create risks, and negatively impact on the tourist experience as well as the country's global reputation.

This project proposes an AI-Powered Tour Guide Booking System with Voice and Retrieval-Augmented Generation (RAG) to address these gaps. The system will provide multi-document verification (ID, licenses, police checks, social media validation), ensuring guide authenticity. A privacy-protected workflow will lock tourists' personal details until bookings are confirmed, enhancing security.

A core innovation is the AI-driven matching and recommendation engine powered by RAG, which delivers personalized, multilingual responses and matches tourists to suitable guides based on preferences, history, and real-time conditions. Safety features such as fraud detection and location tracking further strengthen the platform.

The system is technically feasible, using React.js, Node.js, vector databases, and AI/ML frameworks. Economically, it will be sustained through commission-based revenue, guide subscriptions, and partnerships, with recovery expected within 2 to 3 years.

In summary, this project delivers a secure, intelligent, and user-friendly tourism platform that builds trust, enhances personalization, and improves safety. It has the potential to transform Sri Lanka's tourism industry and set new standards for digital tourism worldwide.

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